The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 Apr 94.

INCH-POUND

MIL-S-19500/562B
7 January 1994
SUPERSEDING
MIL-S-19500/562A
8 September 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, FIELD EFFECT TRANSISTOR, P-CHANNEL, SILICON
TYPES 2N6804 AND 2N6806
JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a P-Channel, enhancement-mode, MOSFET, power transistor intended for use in high density power switching applications. Three levels of product assurance are provided for each encapsulated device type as specified in MIL-S-19500. Two levels of product assurance are provided for each unencapsulated device type.

1.2 Physical dimensions. See figure 1, TO-204AA (formerly TO-3), figures 2, 3, and (see 6.4) for JANHC and JANKC die dimensions.

1.3 Maximum ratings. Unless otherwise specified, T_A = +25°C.

Type	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{DS}	V_{DG}	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C	I_S	I_{DM}	T_J and T_{STG}
	W	W	V dc	V dc	V dc	A dc	A dc	A dc	A(pk)	°C
2N6804	75	4	-100	-100	±20	-11.0	-7.0	-11.0	-50	-55 to +150
2N6806	75	4	-200	-200	±20	-6.5	-4.0	-6.5	-28	-55 to +150

1/ Derate linearly 0.6 W/°C for T_C > +25°C:

$$P_T = \frac{T_{Jmax} - T_C}{R_{\theta JC}}$$

2/
$$I_D = \sqrt{\frac{T_{J\ max} - T_C}{(R_{\theta JC} \times R_{DS(on)} \text{ at } T_{J\ max})}}$$

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: NASA/Parts Project Office (NPPO), NASA Goddard Space Flight Center, Code 311.A, Greenbelt, MD 20771-3917, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026l0-19T03:25Z
Check the source to verify that this is the currentversion before use.

1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 V I_D = -1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = -0.25 mA dc		Max I_{DSS1} V_{GS} = 0 V V_{DS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ 1/ V_{GS} = -10 V dc T_J = +25°C at I_{D2}	Max $r_{DS(on)}$ 1/ V_{GS} = -10 V dc T_J = +150°C at I_{D2}	$R_{\theta JC}$ maximum
	V dc	V dc Min	Max	µA dc	ohm	ohm	°C/W
2N6804	-100	-2.0	-4.0	-25	0.30	0.60	1.67
2N6806	-200	-2.0	-4.0	-25	0.80	1.80	1.67

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

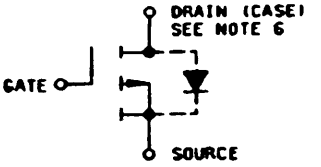
2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

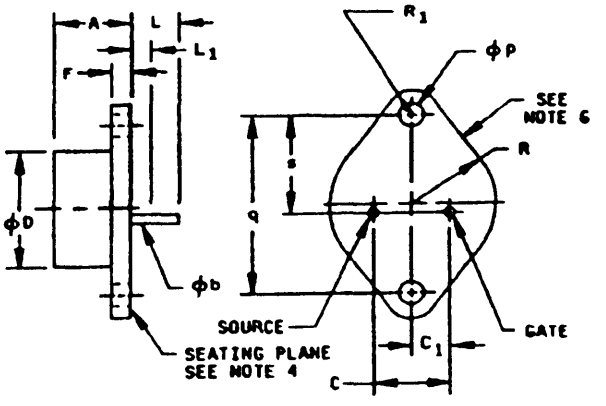
3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.250	.360	6.35	9.15	3
ϕb	.038	.043	0.97	.110	
ϕD		.875		22.23	
C	.420	.440	10.67	11.18	3
C_1	.205	.225	5.21	5.72	3
F	.060	.135	1.52	3.43	
L	.312	.500	7.92	12.70	
L_1		.050		1.27	
ϕp	.151	.161	3.84	4.09	
q	1.177	1.197	29.90	30.40	
R	.495	.525	12.57	13.34	
R_1	.131	.188	3.33	4.78	
s	.655	.675	16.64	17.15	



CIRCUIT SCHEMATIC

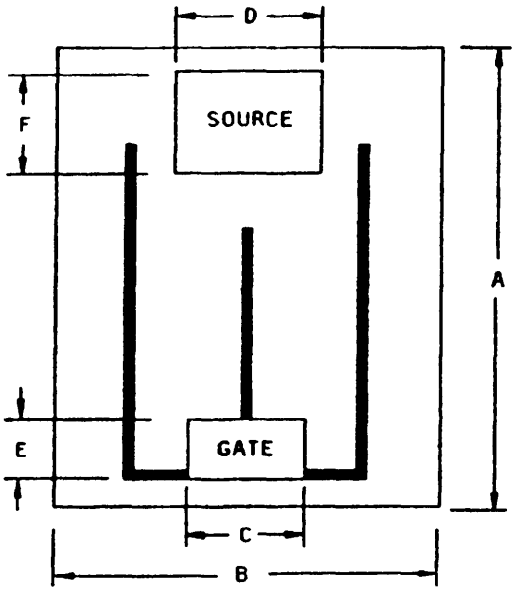


NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) to .055 inch (1.40 mm) below seating plane. When gauge is not used measurement will be made at the seating plane.
4. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
5. Mounting holes shall be deburred on the seating plane side.
6. Drain is electrically connected to case.

FIGURE 1. Physical dimensions of transistor (TO-204AA).

Source: https://assist.dla.mil -- Downloaded: 202610-19T03:25Z
Check the source to verify that this is the current version before use.

2N6804



A-version
HEX-3: -100 V. P-CHANNEL

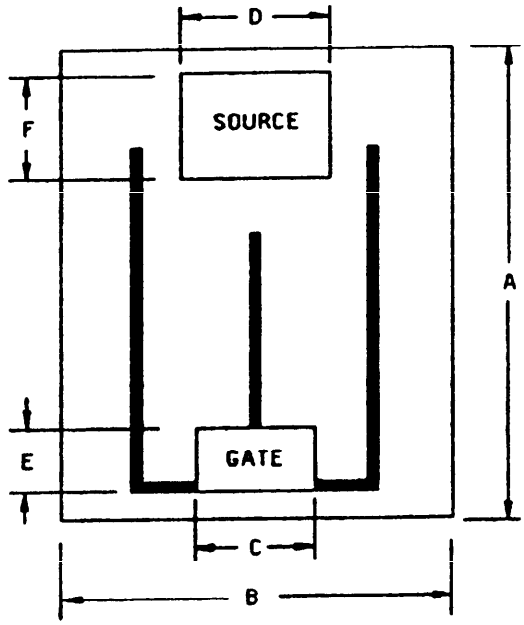
Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	0.172	0.186	4.37	4.72
B	0.106	0.120	2.69	3.05
C	0.021	0.029	0.53	0.74
D	0.035	0.043	0.89	1.09
E	0.014	0.022	0.36	0.56
F	0.025	0.036	0.64	0.91

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are: the back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.
4. The thickness is .0187 inch (0.474 mm), the tolerance is ±.0050 inch (0.13 mm).

FIGURE 2. JANHCA and JANKCA die dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

2N6806



A-version

Dimensions				
Ltr	Inches		Millimeters	
	Min	Max	Min	Max
A	0.173	0.187	4.39	4.75
B	0.108	0.124	2.74	3.15
C	0.022	0.030	0.56	0.76
D	0.030	0.038	0.76	0.97
E	0.012	0.020	0.30	0.51
F	0.021	0.029	0.53	0.74

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die are; the back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.
5. Die thickness is 0.0187 inch (0.475 mm), the tolerance is ±.0050 inch (0.13 mm).

FIGURE 3. JANHCA and JANKCA die dimensions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500 and as follows:

C	Coulomb.
g_{FS}	DC forward transconductance.
IS	Source current through drain (forward biased V_{SD}).
$V_{(BR)DSS}$	Drain to source breakdown voltage, all other terminals short-circuited to source.
$I_{(ISO)}$	Source pin to case isolation current.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500 and on figures 1, 2, and 3 herein.

3.3.1 Lead material and finish. Lead material shall be Kovar, Alloy 52, and a copper core is permitted. Lead finish shall be solderable in accordance with MIL-S-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.3).

3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500. At the option of the manufacturer, marking of country of origin may be omitted from the body of the transistor, but shall be retained on the initial container.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling practices shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500 and table II herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be conducted in accordance with MIL-S-19500 and table II herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

4.2.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with table II of MIL-S-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H. As a minimum, die shall be 100-percent probed in accordance with group A, subgroup 2.

Screen (see table II of MIL-STD-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
2/	Method 3470 (optional)	Method 3470 (optional)
3	Test condition G	Test condition G
9 1/	I_{GSS1}, I_{DSS1}, Subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042, test condition A, t = 240 hours	Method 1042, test condition A
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 µA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)1}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.
2/ Method 3470 is optional if performed as a sample in group A, subgroup 5.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500. Alternate flow is allowed for quality conformance inspection in accordance with figure 2 of MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	1051	Test condition G.
4	1042	Test condition D; 2,000 cycles. The heating cycle shall be 1 minute minimum.
5	1042	Accelerated steady-state operation life; test condition A; V_{DS} = rated T_A = +175°C, t = 120 hours. Read and record $V_{(BR)DSS}$ (pre and post at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10% and I_{DSS} shall not exceed 25μA.
		Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
5	2037	Bond strength (Al-Au die interconnects only); test condition A.
6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
2	1051	Test condition G.
3	1042	Test condition D, 2,000 cycles. The heating cycle shall be 1 minute minimum.
3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	2036	Test condition A; weight = 10 pounds; t = 15 s.
6	1042	Test condition D, 6,000 cycles. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. $R_{\theta JC(max)}$ = 1.67°C/W.

a.	Measuring current (I_M)	10 mA.
b.	Drain heating current (I_H)	4 A.
c.	Heating time (t_H)	Steady-state (see MIL-STD-750, method 3161 for definition).
d.	Drain-source heating voltage (V_H)	25 V.
e.	Measurement time delay (t_{MD})	10 to 60 µs.
f.	Sample window time (t_{SW})	10 µs maximum.

4.5.3 Thermal impedance ($Z_{\theta JC}$ measurements). The $Z_{\theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 3, thermal impedance cures and the group A, subgroup 2 limits) for $Z_{\theta JC}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable $\bar{X}$, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an inline process monitor.

a.	Measuring current (I_M)	10 mA.
b.	Drain heating current (I_H)	4 A minimum.
c.	Heating time (t_H)	100 ms.
d.	Drain-source heating voltage (V_H)	25 V minimum.
e.	Measurement time delay (t_{MD})	30 to 60 µs.
f.	t_{SW} sample window time	10 µs (maximum).

4.5.4 Unclamped inductive switching.

a.	Peak current (I_D)	Rated I_{D1}.
b.	Peak gate voltage (V_{GS})	-10 V.
c.	Gate to source resistor (R_{GS})	25 Ω ≤ R_{GS} ≤ 200 Ω.
d.	Initial case temperature (T_C)	+25°C +10°C, -5°C.
e.	Inductance (L)	100 µH ±10 percent.
f.	Number of pulses to be applied	1 pulse minimum.
g.	Pulse repetition rate	None.

4.5.5 Gate stress test.

V_{GS} = ±30 V minimum.

t = 250 µs minimum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.5.3	$Z_{\theta JC}$		1.2	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V; I_D = -1.0 mA dc Bias condition C	$V_{(BR)DSS}$			
2N6804				-100		V dc
2N6806				-200		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = -0.25 mA dc	$V_{GS(th)1}$	-2.0	-4.0	V dc
Gate reverse current	3411	Bias condition C; V_{DS} = 0 V V_{GS} = +20 and -20 V dc	I_{GSS1}		±100	nA dc
Drain current	3413	V_{GS} = 0; bias condition C V_{DS} = 0 V; V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		-25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = -10 V dc; condition A pulsed (see 4.5.1)	$r_{DS(on)1}$			
2N6804		I_D = -7 A dc			0.30	Ω
2N6806		I_D = -4 A dc			0.80	Ω
Drain to source on-state resistance	3421	V_{GS} = -10 V dc; condition A pulsed (see 4.5.1)	$r_{DS(on)2}$			
2N6804		I_D = -11 A dc			0.36	Ω
2N6806		I_D = -6.5 A dc			0.94	Ω
Forward voltage (source drain diode)	4011	Pulsed (see 4.5.1) V_{GS} = 0 V For devices with a multiple diode structure	V_{SD}			
2N6804		I_S = -11 A dc			-4.7	V
2N6806		I_S = -6.5 A dc			-6.0	V

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		$T_C = T_J = +125°C$				
Gate reverse current	3411	Bias condition C V_{DS} = 0 V V_{GS} = +20 V dc and -20 V dc	I_{GSS2}		±200	nA dc
Drain current	3413	Bias condition C V_{GS} = 0 V V_{DS} = 80 percent rated V_{DS}	I_{DSS2}		-.25	mA dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = -.25 mA	$V_{GS(th)2}$	-1.0		V dc
Static drain to source on-state resistance	3421	V_{GS} = -10 V dc Pulsed (see 4.5.1)	$r_{DS(on)3}$			
2N6804		I_D = -7 A dc			0.55	ohms
2N6806		I_D = -4 A dc			1.60	ohms
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = -.25 mA	$V_{GS(th)3}$		-5.0	V dc
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2} (see 1.3) V_{GS} = -10 V dc R_g 7.5 ohms				
Turn-on delay time			$t_{d(on)}$			
2N6804		V_{DD} = -35 V dc			60	ns
2N6806		V_{DD} = -63 V dc			50	ns
Rise time			t_r			
2N6804		V_{DD} = -35 V dc			140	ns
2N6806		V_{DD} = -63 V dc			100	ns

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued						
Turn-off delay time			$t_{d(off)}$			
2N6804		V_{DD} = -35 V dc			140	ns
2N6806		V_{DD} = -63 V dc			100	ns
Fall time			t_f			
2N6804		V_{DD} = -35 V dc			140	ns
2N6806		V_{DD} = -63 V dc			80	ns
Subgroup 5						
Single pulse unclamped inductive switching 3/	3470	See 4.5.4, 116 devices, c = 0				
Electrical measurements		See table I, group A, subgroup 2.				
Safe operating area test	3474	See figure 5 V_{DS} = 80 percent of rated V_{DS} V_{DS} ≤ 200 V maximum				
		t_p = 10 ms				
Electrical measurements		See table I, group A, Subgroup 2.				
Subgroup 6						
Not applicable						
Subgroup 7						
Gate charge	3471	Condition B				
Test 1						
On-state gate charge			$Q_{g(on)}$			
2N6804					29.0	nC
2N6806					31.0	nC
Test 2						
Gate to source charge			Q_{gs}			
2N6804					7.1	nC
2N6806					7.0	nC

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7 - Continued						
Test 3						
Gate to drain charge			Q_{gd}			
2N6804					21.0	nC
2N6806					17.0	nC
Reverse recovery time	3473	$V_{DD} \leq -50$ V	t_{rr}			
2N6804		di/dt ≤ -100 A/µs			250	ns
2N6806		$I_F = -11$ A di/dt ≤ -100 A/µs $I_F = -6.5$ A			400	ns

1/ For sampling plan, see MIL-S-19500.

2/ This test is required for the following endpoint measurements only (not intended for screen 13):

JANS - group B, subgroups 3 and 4.

JANTX and JANTXV - group B, subgroups 2 and 3,
group C, subgroup 6,
group E, subgroup 1.

3/ This test is optional if performed as a 100% screen.

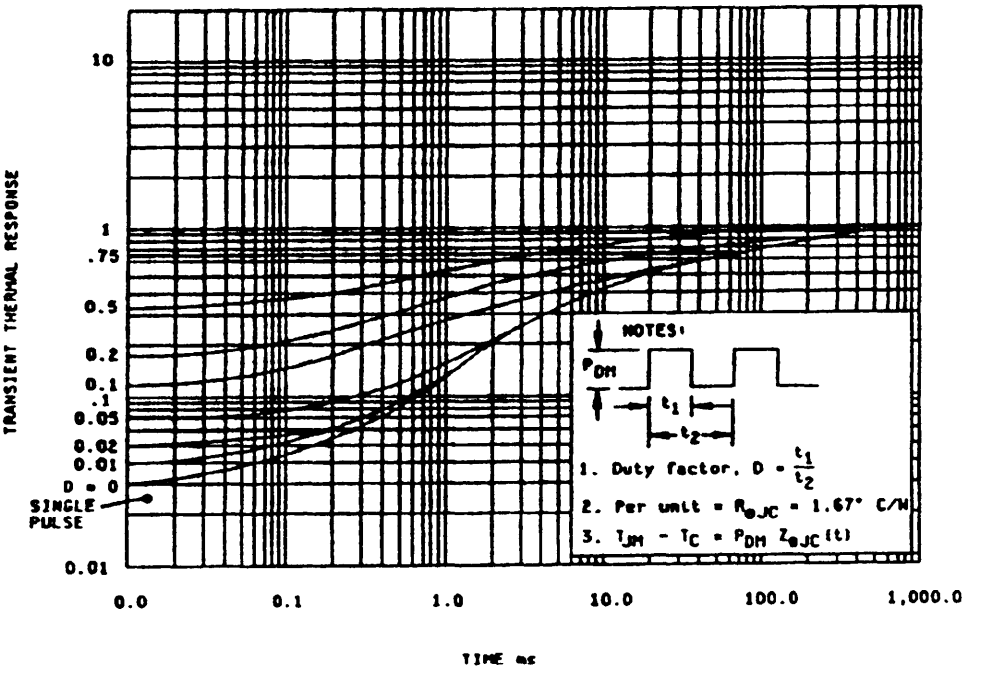
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.

TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection 1/	MIL-STD-750		Qualification conformance inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 0
Thermal shock (temperature cycling)	1051	Condition G, 500 cycles	
Electrical measurements		See table I, group A, subgroup 2.	
Subgroup 2 2/			45 devices, c = 0
Steady-state reverse bias	1042	Condition A; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2.	
Steady-state gate bias	1042	Condition B; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2.	
Subgroup 3			
Not applicable			
Subgroup 4			5 devices, c = 0
Thermal resistance	3161	$R_{\theta JC}$ = 1.67°C/W maximum, see 4.5.2	
Subgroup 5			
Not applicable			

1/ JANHC and JANKC devices are qualified in accordance with appendix H of MIL-S-19500.
2/ A separate sample may be pulled for each test.

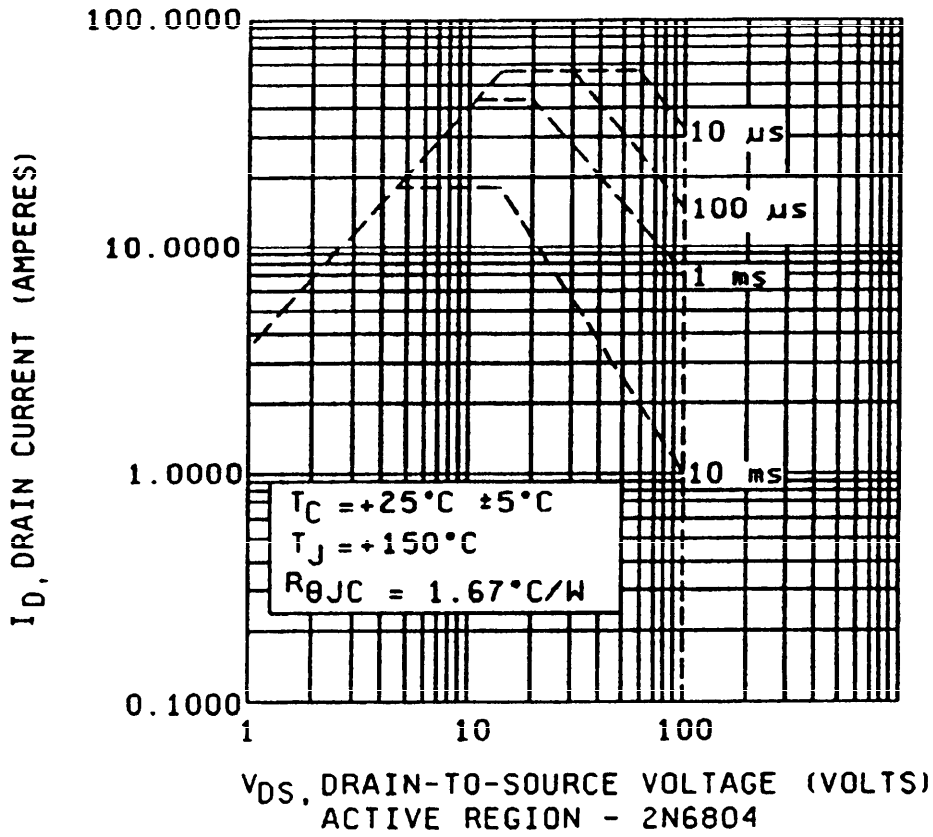
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.



NOTE: These curves refer to devices packaged in a TO-204AA package only.

FIGURE 4. Transient thermal impedance.

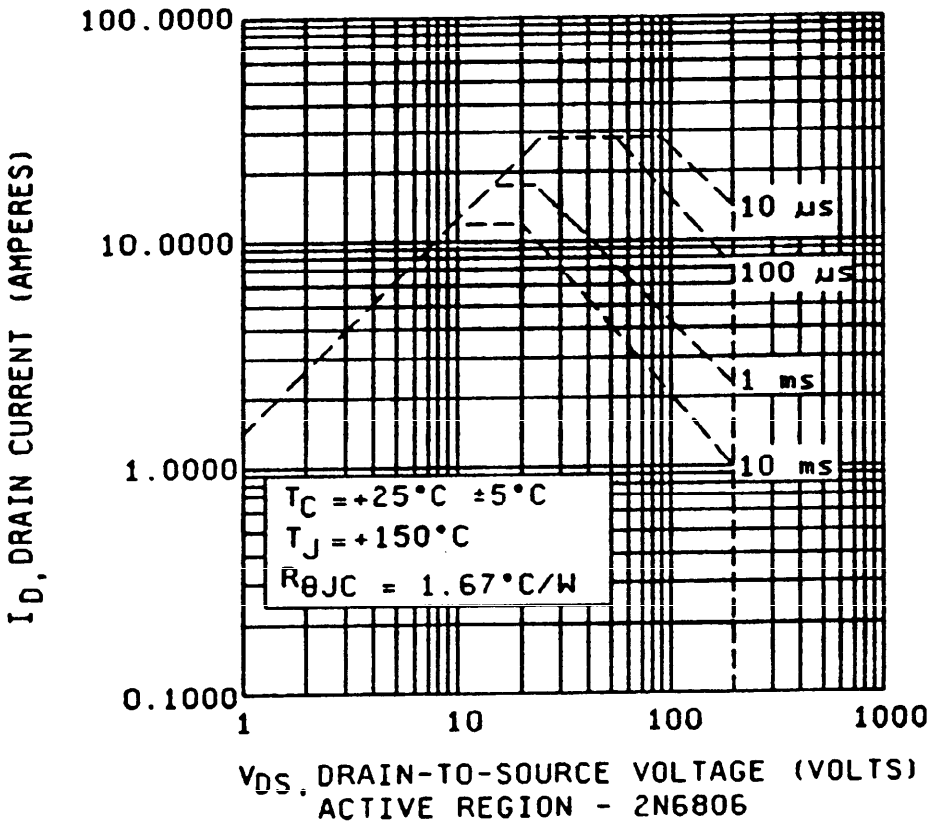
Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.



NOTE: These curves refer to devices packaged in a TO-204AA package only.

FIGURE 5. Maximum safe operating area.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.



NOTE: These curves refer to devices packaged in a TO-204AA package only.

FIGURE 5. Maximum safe operating area - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T03:25Z
Check the source to verify that this is the current version before use.

generator source impedance shall be 50Ω.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Cross-reference and complement list. Parts from this specification may be used to replace the following commercial Part or Identifying Number (PIN's). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification.

Preferred types	Commercial types	Complement
2N6804	IRFF9130, IRFF9131, IRFF9132, IRFF9133	2N6756
2N6806	IRFF9230, IRFF9231, IRFF9232, IRFF9233	2N6758

6.3 Acquisition requirements. Acquisition documents must specify the following:

a. Issue of DODISS to be cited in the solicitation (see 2.1.1 and 2.2).

b. The lead finish as specified (see 3.3.1).

c. For die acquisition, the JANHC or JANKC letter version shall be specified (see figures 2 and 3).

d. Type designation and quality assurance level.

6.4 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example JANHCA2N6804) will be identified on the QPL.

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Navy - TD
Air Force - 19, 70, 80
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 5961-1557)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T03:25Z
Check the source to verify that this is the current version before use.